

St George Girls High School

Year 12

Common Test 3

May 2006



Mathematics Extension 2

Time Allowed: 75 Minutes

Instructions to Candidates

1. Both questions may be attempted.
2. All necessary working must be shown.
3. Start each question on a new page.

Question 1 – (25 marks) – Start a New Page

Marks

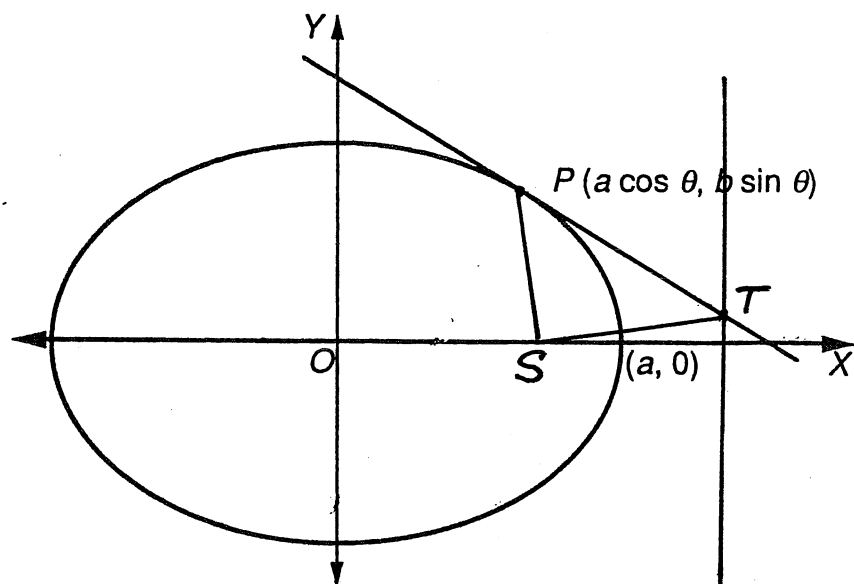
- a) Show that the curves $x^2 - y^2 = c^2$ and $xy = c^2$ intersect at right angles.
Do not attempt to solve these equations simultaneously. 3
- b) For the hyperbola $xy = 16$, write down the:
- (i) eccentricity 1
 - (ii) coordinates of the vertices 2
 - (iii) coordinates of the foci 2
 - (iv) equations of the directrices 2
- c) (i) Prove that the equation of the tangent to the curve $xy = c^2$ at the point $P\left(ct, \frac{c}{t}\right)$ is $x + t^2y = 2ct$. 2
- (ii) Hence, prove that for the rectangular hyperbola $xy = c^2$, the area of the triangle bounded by a tangent and the asymptotes is a constant. 3
- d) P and Q are the points on the rectangular hyperbola $H : xy = c^2$ with parameters p and q respectively ($p > 0, q > 0$).
- (i) Find the point of intersection, T , of the tangents to H at P and Q (you may use the result from c(i)). 2
 - (ii) Find the locus of T as P and Q vary if $pq = k$ (k constant). 3

Question 1 (cont'd)

Marks

e)

5



The tangent to the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $P(a \cos \theta, b \sin \theta)$ meets the directrix at T as shown above. If S is the corresponding focus, show that PT subtends a right angle at S .

[NOTE: The equation of the tangent to E at P is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$]

Question 2 – (25 marks) – Start a new page

Marks

a) The roots of $x^3 + 2x - 1 = 0$ are α , β and γ .

(i) Find the polynomial equation with roots:

(α) $-\alpha, -\beta, -\gamma$

1

(β) $\alpha, -\alpha, \beta, -\beta, \gamma, -\gamma$

1

(γ) $\alpha^2, \beta^2, \gamma^2$

2

(δ) $\alpha - 1, \beta - 1, \gamma - 1$

1

(ii) Show the $(\alpha + \beta - \gamma)(\beta + \gamma - \alpha)(\gamma + \alpha - \beta) = -8$

3

b) It is given that $\cos 4\theta = 8\cos^4 \theta - 8\cos^2 \theta + 1$

(i) Use $x = \cos \theta$ to find the solutions of $16x^4 - 16x^2 + 1 = 0$

3

(ii) Hence evaluate $\cos \frac{\pi}{12}$ and $\cos \frac{5\pi}{12}$ leaving the answers in surd form.

6

c) Consider the polynomial equation $x^3 + ax + b = 0$

(i) Show that if $a > 0$ then the equation has only one real root.

2

(ii) What condition must 'a' and 'b' satisfy if the equation has three real roots?

6

QUESTION I:

$$(a) \quad x^2 - y^2 = c$$

$$\Rightarrow \frac{d}{dx}(x^2) - \frac{d}{dx}(y^2) = 0$$

$$\therefore 2x - 2y \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = \frac{x}{y} \quad \text{--- (1)}$$

If (x_1, y_1) is point of intersection then

$$(1) \Rightarrow \frac{dy}{dx} = \frac{x_1}{y_1} = m_1$$

$$\text{also } xy = c$$

$$\Rightarrow y = c x^{-1}$$

$$\therefore \frac{dy}{dx} = -\frac{c}{x^2}$$

$$\text{at } (x_1, y_1) \quad \frac{dy}{dx} = -\frac{c}{x_1^2} = m_2$$

$$\text{Then } m_1 \times m_2 = \frac{x_1}{y_1} \times -\frac{c}{x_1^2}$$

$$= -\frac{c}{x_1 y_1}$$

$$= -\frac{c}{c} \text{ since } (x_1, y_1) \text{ lies on } xy = c$$

$$= -1$$

$\therefore$ At points of intersection the curves are perpendicular.

(b)

$$xy = 16$$
$$= \frac{1}{2} a^2$$

$$\Rightarrow a^2 = 32$$

$$\therefore a = 4\sqrt{2} \quad (a > 0)$$

(i) $e = \sqrt{2}$

(ii) Vertices are at $(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}})$ and $(-\frac{a}{\sqrt{2}}, -\frac{a}{\sqrt{2}})$

$$= (4, 4) \text{ and } (-4, -4)$$

(iii) Foci are at (a, a) and $(-a, -a)$

$$= (4\sqrt{2}, 4\sqrt{2}) \text{ and } (-4\sqrt{2}, -4\sqrt{2})$$

(iv) Directrices are $x + y = a$, $x + y = -a$

ie $x + y = 4\sqrt{2}$, $x + y = -4\sqrt{2}$

(c)

$$xy = c^2$$
$$\Rightarrow y = c^2 x^{-1}$$
$$\frac{dy}{dx} = -c^2 x^{-2}$$
$$= -\frac{c^2}{x^2}$$

at $P(ct, \frac{c}{t})$ $\frac{dy}{dx} = -\frac{c^2}{c^2 t^2}$

$$= -\frac{1}{t^2}$$

$\therefore$ Tangent at P is :

$$y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$$

$$t^2 y - ct = -x + ct$$

ie $x + t^2 y = 2ct$

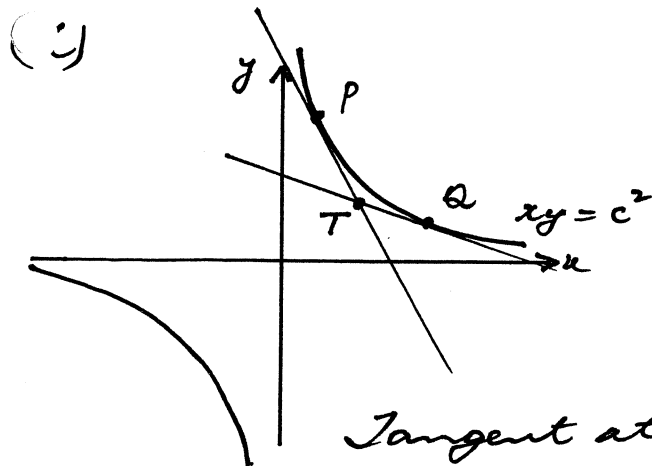
(ii) Tangent at P crosses
 x axis at $A(2ct, 0)$
 y axis at $B(0, \frac{2c}{t})$

$\therefore$ Area of triangle ABO is

$$\frac{1}{2} \cdot 2ct \cdot \frac{2c}{t}$$

$$= 2c^2$$

$$= \text{constant.}$$



$$\text{Tangent at P: } x + p^2 y = 2cp \quad \text{--- (1)}$$

$$\text{Tangent at Q: } x + q^2 y = 2cq \quad \text{--- (2)}$$

$$\text{(1) - (2): } y(p^2 - q^2) = 2c(p - q)$$

$$y(p+q)(p-q) = 2c(p-q)$$

$$\therefore y = \frac{2c}{p+q}, \quad p \neq q$$

sub in (1)

$$\therefore x + \frac{2cp^2}{p+q} = 2cp$$

$$x = 2cp - \frac{2cp^2}{p+q}$$

$$= \frac{2cp^2 + 2cpq - 2cp^2}{p+q}$$

$$= \frac{2cpq}{p+q}$$

$$\therefore T \equiv \left(\frac{2cpq}{p+q}, \frac{2c}{p+q} \right)$$

(ii) If $pq = k$ (constant)

$$\text{Then } T \Rightarrow \left(\frac{2ck}{p+q}, \frac{2c}{p+q} \right)$$

$$\text{ie } x = \frac{2ck}{p+q}, \quad y = \frac{2c}{p+q}$$

$$\therefore \frac{y}{x} = \frac{\frac{2c}{p+q}}{\frac{2ck}{p+q}}$$

$$= \frac{1}{k}$$

$$\text{ie } y = \left(\frac{1}{k} \right) x$$

$$\text{sub } y = \left(\frac{1}{k} \right) x \text{ in } xy = c^2$$

$$\therefore x \cdot \frac{x}{k} = c^2$$

$$x^2 = c^2 k$$

$$x = c\sqrt{k} \quad (x > 0 \text{ since } p > 0, q > 0)$$

$\therefore$ Locus of T is the line $y = \left(\frac{1}{k} \right) x$

with $0 < x < c\sqrt{k}$

(since $\frac{2ck}{p+q} \neq 0$)

(e) Tangent at P is:

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

Tangent intersects directrix $x = \frac{a}{e}$ at $T(\frac{a}{e}, y_1)$

$$\therefore \frac{\cos \theta}{e} + \frac{y_1 \sin \theta}{b} = 1$$

$$b \cos \theta + y_1 \cdot e \sin \theta = be$$

$$\therefore y_1 = \frac{be - b \cos \theta}{e \sin \theta}$$

$$\therefore T \equiv \left(\frac{a}{e}, \frac{be - b \cos \theta}{e \sin \theta} \right)$$

$$m_{TS} = \frac{\frac{be - b \cos \theta}{e \sin \theta} - 0}{\frac{a}{e} - ae}$$

$$m_{PS} = \frac{b \sin \theta - 0}{a \cos \theta - ae}$$

$$\begin{aligned} \therefore m_{TS} \times m_{PS} &= \left(\frac{\frac{be - b \cos \theta}{e \sin \theta}}{\left(\frac{a - ae^2}{e} \right)} \right) \times \frac{b \sin \theta}{a(\cos \theta - e)} \\ &= \frac{b(e - \cos \theta)}{\cancel{e \sin \theta}} \times \frac{\cancel{e}}{a(1 - e^2)} \times \frac{b \sin \theta}{a(\cos \theta - e)} \\ &= \frac{b^2(e - \cos \theta)}{a^2(1 - e^2) \cdot (\cos \theta - e)} \\ &= -1 \text{ since } b^2 = a^2(1 - e^2). \end{aligned}$$

∴ m_{TS} × m_{PS} = -1 ∴ TS ⊥ PS ∴ P is on the circle with TS as diameter.

QUESTION 2:

$$(a) \quad (i) \quad (2) \quad P(-x) = 0$$

$$\Rightarrow (-x)^3 + 2(-x) - 1 = 0$$

$$\text{i.e. } -x^3 - 2x - 1 = 0$$

$$\text{i.e. } x^3 + 2x + 1 = 0$$

$$(3) \quad (x^3 + 2x + 1)(x^3 + 2x - 1) = 0$$

$$\text{i.e. } x^6 + 2x^4 - x^3 + 2x^4 + 4x^2 - 2x + x^3 + 2x - 1 = 0$$

$$\text{i.e. } x^6 + 4x^4 + 4x^2 - 1 = 0$$

$$(4) \quad P(\sqrt{x}) = 0$$

$$\text{i.e. } (\sqrt{x})^3 + 2(\sqrt{x}) - 1 = 0$$

$$x\sqrt{x} + 2\sqrt{x} = 1$$

$$\sqrt{x}(x+2) = 1$$

squaring

$$\Rightarrow x(x+2)^2 = 1$$

$$x(x^2 + 4x + 4) - 1 = 0$$

$$\text{i.e. } x^3 + 4x^2 + 4x - 1 = 0$$

$$(5) \quad P(x+1) = 0$$

$$(x+1)^3 + 2(x+1) - 1 = 0$$

$$x^3 + 3x^2 + 3x + 1 + 2x + 2 - 1 = 0$$

$$\text{i.e. } x^3 + 3x^2 + 5x + 2 = 0$$

(ii) Roots of $x^3 + 2x - 1 = 0$ are α, β, γ

$$\therefore \alpha + \beta + \gamma = 0 \quad \text{--- (1)}$$

$$\text{Then } (\alpha + \beta - \gamma)(\beta + \gamma - \alpha)(\gamma + \alpha - \beta)$$

$$= (-\gamma - \gamma)(-\alpha - \alpha)(-\beta - \beta) \text{ from (1)}$$

$$= (-2\gamma)(-2\alpha)(-2\beta)$$

$$= -8\alpha\beta\gamma$$

$$0 \quad \therefore \alpha\beta\gamma = 1$$

$$(b) \quad \cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1 \quad \text{--- (1)}$$

$$(i) \quad 16x^4 - 16x^2 + 1 = 0 \quad \text{--- (2)}$$

$$\text{let } x = \cos \theta$$

$$\Rightarrow 16\cos^4\theta - 16\cos^2\theta + 1 = 0$$

$$\Rightarrow 8\cos^4\theta - 8\cos^2\theta = -\frac{1}{2}$$

$$\Rightarrow \cos 4\theta - 1 = -\frac{1}{2}$$

$$\therefore \cos 4\theta = \frac{1}{2}$$

$$\therefore 4\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}, \frac{17\pi}{3}, \dots$$

$$\therefore \theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}, \dots$$

$$\therefore x = \cos \frac{\pi}{12}, \cos \frac{5\pi}{12}, \cos \frac{7\pi}{12}, \cos \frac{11\pi}{12}, \dots$$

$$= \cos \frac{\pi}{12}, \cos \frac{5\pi}{12}, -\cos \frac{5\pi}{12}, -\cos \frac{\pi}{12}$$

$$(ii) \text{ From (2): } x^2 = \frac{16 \pm \sqrt{192}}{32}$$

$$= \frac{16 \pm 8\sqrt{3}}{32}$$

$$= \frac{2 \pm \sqrt{3}}{4}$$

$$= \frac{2-\sqrt{3}}{4}, \frac{2+\sqrt{3}}{4}$$

Since $\cos \frac{\pi}{12} > \cos \frac{5\pi}{12}$ it follows that

$$\cos \frac{\pi}{12} = \left(\frac{2+\sqrt{3}}{4} \right)^{\frac{1}{2}} \quad \cos \frac{5\pi}{12} = \left(\frac{2-\sqrt{3}}{4} \right)^{\frac{1}{2}}$$

$$(c) \quad p(x) = x^3 + ax + b = 0$$

$$(i) \quad p'(x) = 3x^2 + a$$

stationary points when $p'(x) = 0$

$$\text{ie } 3x^2 = -a$$

$$x^2 = -\frac{a}{3}$$

If $a > 0$ Then there are NO stationary [1]
points.

BUT $p(x) \rightarrow \infty$ as $x \rightarrow \infty$

and $p(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ [1]

$\therefore$ There must be one value of x for which $p(x) = 0$

(ii) Three real roots

$\Rightarrow$ 2 stationary points whose y -values have opposite signs. [1]

$$p'(x) = 0 \Rightarrow x^2 = -\frac{a}{3}$$

$$\therefore x_1 = \left(-\frac{a}{3}\right)^{\frac{1}{2}}$$

$$x_2 = -\left(-\frac{a}{3}\right)^{\frac{1}{2}}$$

$$p(x_1) = \left(-\frac{a}{3}\right)^{\frac{3}{2}} + a\left(-\frac{a}{3}\right)^{\frac{1}{2}} + b$$

$$= \left(-\frac{a}{3}\right)^{\frac{1}{2}} \left[-\frac{a}{3} + a\right] + b$$

$$= \left(-\frac{a}{3}\right)^{\frac{1}{2}} \left(\frac{2a}{3}\right) + b$$

[1]

$$P(x_2) = -\left(-\frac{a}{3}\right)^{\frac{3}{2}} - a\left(-\frac{a}{3}\right)^{\frac{1}{2}} + b$$

$$= \left(-\frac{a}{3}\right)^{\frac{1}{2}} \left[\frac{a}{3} - a\right] + b$$

$$= \left(-\frac{2a}{3}\right) \left(-\frac{a}{3}\right)^{\frac{1}{2}} + b \quad [1]$$

we must have $P(x_1) \cdot P(x_2) < 0$

$$\text{ie } \left[b + \frac{2a}{3} \left(-\frac{a}{3}\right)^{\frac{1}{2}}\right] \left[b - \left(\frac{2a}{3}\right) \left(-\frac{a}{3}\right)^{\frac{1}{2}}\right] < 0 \quad [1]$$

$$\text{ie } b^2 - \frac{4a^2}{9} \left(-\frac{a}{3}\right) < 0$$

$$27b^2 + 4a^3 < 0 \quad [1]$$
